where m_{nucl} is the nuclear mass. For the light hydrogen isotope, the H_α line (i.e. the transition energy between $n_1 = 1$ and $n_2 = 2$) occurs at 656.285 nm, while the corresponding line for the deuterium isotope of hydrogen occurs at 656.106 nm. This difference could be predicted from (3.23) and its observation experimentally in 1932 provided the first evidence for the existence of a heavy hydrogen isotope. This spectral difference has practical importance as it can be used for a spectroscopic analysis of the amount of heavy water in light water. Similar *isotopic line shifts* occur in the spectra of all elements, but are rarely as large as the shift of almost 0.2 nm observed for hydrogen. For the isotopes ^{235}U and ^{238}U, the isotopic shift is 0.025 nm.

3.6. ISOTOPE EFFECTS IN CHEMICAL EQUILIBRIUM

In a mixture of molecules AX and BX, with a common element X, an exchange of the atoms of the common element between the two molecules may occur. When the two compounds have different isotopes X and X*, we may have an *isotope exchange* according to

$$AX + BX^* = AX^* + BX \tag{3.25}$$

The equilibrium constant k in the reaction is given by

$$\Delta G^{\circ} = -\mathbf{R}T \ln k = -\mathbf{R}T \ln \{([AX^*][BX])/([AX][BX^*])\} \tag{3.26}$$

where ΔG° is the (Gibb's) free energy and $\mathbf{R}$ is the molar gas constant. For values of fundamental constants see Appendix III.

3.6.1. The partition function

It has been shown that k deviates slightly from 1, giving rise to the isotopic effects observed in nature for the light elements. This deviation can be calculated by methods of statistical thermodynamics. Only the main features of this theory are given here. The equilibrium constant k can be written

$$k = (F_{AX^*}\, F_{BX})/(F_{AX}\, F_{BX^*}) \tag{3.27}$$

where F is the *grand partition function*, which for each molecule includes all possible energy states of the molecule and the fraction of molecules in each energy state under given external condition. The grand partition function is defined by

$$F = f_{\text{tr}}\, f_{\text{rot}}\, f_{\text{vib}}\, f_{\text{el}}\, f_{\text{nsp}} \tag{3.28}$$

where each term f_j refers to a particular energy form: translation, rotation, vibration, electron movement, and nuclear spin. The two latter will have no influences on the chemical isotope effect, and can therefore be omitted. It can be shown that each separate partition function f_j can be described by the expression

$$f_j = \Sigma g_{j,i} e^{-E\text{j,i}/\mathbf{k}T} \tag{3.29}$$

where $E_{j,i}$ is the particular energy state i for the molecule's energy mode j; e.g. for $j =$ vibration, there may be 20 different vibrational states (i.e. the maximum i-value is 20) populated. $\mathbf{k}$ is the Boltzmann constant; (3.29) is closely related to the Boltzmann distribution law (see next section).

The term $g_{j,i}$ is called the *degeneracy* and corrects for the fact that the same energy state in the molecule may be reached in several different ways. The summation has to be made over all energy states i.

3.6.2. Kinetic energy and temperature

The kinetic energy of one mole of an ideal gas at temperature T (K) is given by its translational energy, which according to the kinetic theory of gases is

$$E_{\text{tr}} = 3\mathbf{R}T/2 \left(\text{J mole}^{-1}\right) \tag{3.30}$$

Dividing by the Avogadro number $\mathbf{N}_{\text{A}}$ yields the average kinetic energy per molecule (or particle)

$$\overline{E}_{\text{tr}} = 3\mathbf{k}T/2 \left(\text{J particle}^{-1}\right) \tag{3.31}$$

where $\mathbf{k} = \mathbf{R}/\mathbf{N}_{\text{A}}$. From mechanics we know that the kinetic energy of a single particle of mass m and velocity v is

$$E_{\text{kin}} = \tfrac{1}{2}mv^2 \tag{3.32}$$

Summing over a large number of particles, we can define an average kinetic energy

$$\overline{E}_{\text{kin}} = \tfrac{1}{2}m\overline{v}^2 \tag{3.33}$$

where $\overline{v}^2$ is the mean square velocity. Because (3.31) is the average kinetic energy at temperature T, it must equal (3.33), *i.e.* $\overline{E}_{\text{tr}} = \overline{E}_{\text{kin}}$, thus

$$3\mathbf{k}T/2 = \tfrac{1}{2}m\overline{v}^2 \tag{3.34}$$

Thus for a given temperature there is a corresponding *average particle velocity*. However, the individual particles are found to move around with slightly different velocities. J. C. Maxwell calculated the velocity distribution of the particles in a gas. For the simplest possible system it is a *Boltzmann distribution*. In a system of n_{o} particles the number of particles n_E that have kinetic energy $> E$ is given by

$$n_E = n_o\left(2/\sqrt{\pi}\right)(\mathbf{k}T)^{-3/2} \int_E^{\infty} \sqrt{E}\,\text{e}^{-E/\mathbf{k}T}\,\text{d}E \tag{3.35}$$

In Fig. 3.4 n_E/n_{o} is plotted as a function of E for three different T's. For the line at 290 K we have marked the energies $\mathbf{k}T$ and $3\mathbf{k}T/2$. While $3\mathbf{k}T/2$ (or rather $3\mathbf{R}T/2$)

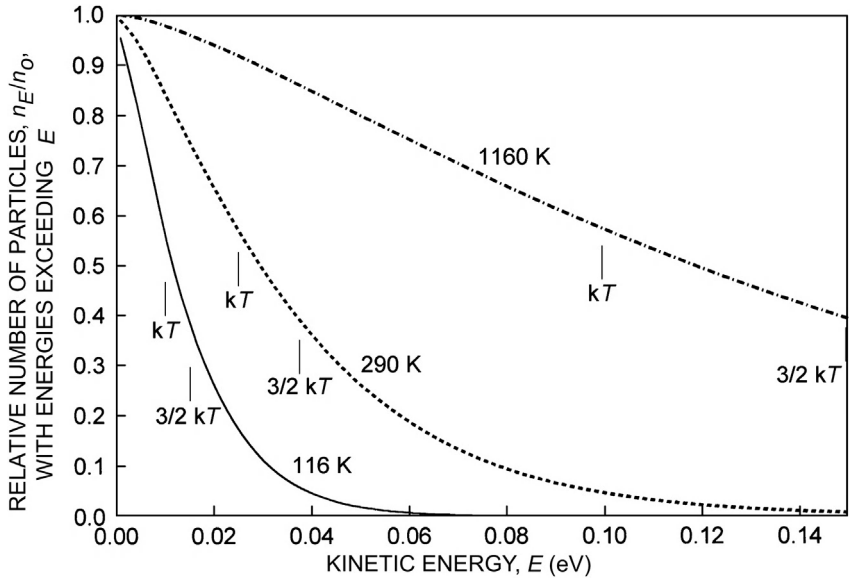


Figure 3.4 Relative number of particles as function of energy.

corresponds to the thermodynamic average translational energy, $\mathbf{k}T$ corresponds to the *most probable kinetic energy*: the area under the curve is divided in two equal halves by the $\mathbf{k}T$ line.

In chemistry, the thermodynamic energy (3.31) must be used, while in nuclear reactions the most probable energy E' must be used,

$$E' = \tfrac{1}{2} m(v')^2 = \mathbf{k}T \tag{3.36}$$

where v' is the *most probable velocity.*

Although the difference between $\overline{E}$ and E' is not large (*e.g.* at 17°C = 0.037 eV and $E' = 0.025$ eV), it increases with temperature. *The most probable velocity is the deciding factor whether a nuclear reaction takes place or not*. Using (3.35) we can calculate the fraction of particles at temperature T having a higher energy than $\mathbf{k}T$: $E > 2\mathbf{k}T$, 26%, $> 5\mathbf{k}T$; 1.86%, and $> 10\mathbf{k}T$, 0.029%. This high-energy tail is of importance for chemical reaction yields because it supplies the molecules with energies in excess of the activation energy. It is also of importance for nuclear reactions and in particular for thermonuclear processes.

3.6.3. The partial partition functions

So far there has been no clue to why isotopic molecules such as H_2O and D_2O, or $H^{35}Cl$ and $H^{37}Cl$, behave chemically in slightly different manner. This explanation comes from a study of the energy term E_j, which contains the atomic masses for all three molecular movements: translation, rotation, and vibration.

(a) *Translational energy.* The translational energy, as used in chemical thermodynamics involves molecular movements in all directions of space. The energy is given by the expression (3.31). A more rigorous treatment leads to the expression

$$f_{\mathrm{tr}} = (2\pi M\mathbf{k}T)^{3/2} V_{\mathrm{M}}\mathbf{h}^{-3} \tag{3.37}$$

for the translational partition function, where V_{M} is the molar volume and **h** the Planck constant. Notice that no quantum numbers appear in (3.37); the reason is that they are not known because of the very small ΔE's of such changes.

(b) *Rotational energy.* Taking the simplest case, a linear diatomic molecule with atomic masses m_1 and m_2 at a distance r apart, the rotational energy is given by

$$E_{\mathrm{rot}} = I_{\mathrm{rot}}\boldsymbol{\omega}^2 \tag{3.38}$$

where I_{rot} is the rotational moment of inertia and ω the angular velocity of rotation (radians s^{-1}). I_{rot} is calculated from

$$I_{\mathrm{rot}} = m_{\mathrm{red}}r^2 \tag{3.39}$$

where the reduced mass $m_{\mathrm{red}} = (m_1^{-1} + m_2^{-1})^{-1}$. Equation (3.38), derived from classical mechanics, has to be modified to take into account that only certain energy states are permitted

$$E_{\mathrm{rot}} = \mathbf{h}^2 n_{\mathrm{r}}(n_{\mathrm{r}} + 1)/\left(8\pi^2 I_{\mathrm{rot}}\right) \tag{3.40}$$

where n_{r} is the rotational quantum number. For transformation of energy in eV to the wave number $\overline{\nu}$ or wavelength λ of the corresponding photon energy, the relation

$$\Delta E\,(\mathrm{eV}) = 1.23980 \times 10^{-4}\overline{\nu}\left(\mathrm{cm}^{-1}\right) \tag{3.41}$$

is used, where $\overline{\nu} = 1/\lambda$. (For blue light of about 480 nm the following relations are obtained: 480 nm $= 4.8 \times 10^{-7}$ m $= 0.48$ μm $= 20\,833$ $\mathrm{cm}^{-1} = 2.58$ eV.) The rotational energies are normally in the range $0.001 - 0.1$ eV, i.e. the wavelength region $10^{-3} - 10^{-5}$ m. The partition function for the rotational energy is obtained by introducing (3.40) into (3.29). More complicated expressions are obtained for polyatomic and nonlinear molecules.

(c) *Vibrational energy.* For a diatomic molecule the vibrational energy is given by

$$E_{\mathrm{vib}} = \mathbf{hc}\boldsymbol{\omega}_{\mathrm{v}}\left(n_{\mathrm{v}} + \tfrac{1}{2}\right) \tag{3.42}$$

where

$$\boldsymbol{\omega}_{\mathrm{v}} = \tfrac{1}{2}(k'/m_{\mathrm{red}})^{1/2}/(\pi\mathbf{c}) \tag{3.43}$$

is the zero point vibrational frequency (the molecule is still vibrating at absolute zero, when no other movements occur) and n_{v} the vibrational quantum number. k' is the *force constant* for the particular molecule.

3.6.4. The isotopic ratio

It is seen from this digression that the mass of the molecular atoms enters into the partition functions for all three modes of molecular movement. The largest energy changes are associated with the vibrational mode, and the isotopic differences are also most pronounced here. Neglecting quantization of translational and rotational movements, one can show that

$$F = f_{\mathrm{tr}} f_{\mathrm{rot}} f_{\mathrm{vib}} = 2\left(2\pi M\mathbf{k}T/\mathbf{h}^3\right)^{3/2} V_{\mathrm{M}} 8\pi^2 I_{\mathrm{r}}\mathbf{k}T\,\mathrm{e}^{-u/2}/\left\{\mathbf{h}^2 k_{\mathrm{s}}\left(1-\mathrm{e}^{-u}\right)\right\} \tag{3.44}$$

where k_{s} is a symmetry constant for rotation and $u = \mathbf{hc}\omega/\mathbf{k}T$. This expression holds for all molecules in (3.25). Thus for the ratio one gets

$$F_{\mathrm{AX}}/F_{\mathrm{AX}^*} = (M^*/M)^{3/2} k_{\mathrm{s}} I_{\mathrm{r}}^* \left(1-\mathrm{e}^{-u}\right)\mathrm{e}^{-u^*/2}/\left\{k_{\mathrm{s}}^* I_{\mathrm{r}}\left(1-\mathrm{e}^{-u^*}\right)\mathrm{e}^{-u/2}\right\} \tag{3.45}$$

where the asterisk refers to molecule AX*. This relation shows the mass dependency of the equilibrium constant (3.26) (a similar relation holds for the BX-BX* combination). (3.45) contains factors all of which can be determined spectroscopically. Consequently, the equilibrium constants for isotope exchange reactions can be, and have been, calculated for many systems. We see here, as commented earlier that as for the physical effects of isotopic variations also the chemical one is highly dependent on the percentage change. Thus, also here we expect larger effects for lighter molecules.

3.6.5. Paleotemperatures and other applications

Eqn (3.45) contains the temperature in the exponent. Isotope exchange equilibria are thus temperature dependent. A practical use of this fact is the determination of paleotemperatures. In the isotope exchange reaction

$$\mathrm{C}\,{}^{16}\mathrm{O}_2(\mathrm{g}) + \mathrm{H}_2\,{}^{18}\mathrm{O}\,(\mathrm{l}) = \mathrm{C}\,{}^{18}\mathrm{O}\,{}^{16}\mathrm{O}(\mathrm{g}) + \mathrm{H}_2\,{}^{16}\mathrm{O}(\mathrm{l}) \tag{3.46}$$

the oxygen isotopes in CO_2 exchange with the oxygen isotopes in H_2O. The value of the equilibrium constant (mole fractions) $k = 1.046$ (0°C) indicates that ^{18}O will be slightly enriched in the CO_2 by the reaction. Thus if carbon dioxide from combustion is bubbled through water, the emergent gas will be more enriched in the ^{18}O than the residual water. In this reaction, the *isotope effect* is said to be 4.6%. The following reaction occurs with carbonate ions in water:

$$\mathrm{Ca}^{2+} + \mathrm{C}\,{}^{16}\mathrm{O}_3{}^{2-} + \mathrm{H}_2\,{}^{18}\mathrm{O} = \mathrm{CaC}\,{}^{18}\mathrm{O}\,{}^{16}\mathrm{O}_2(\mathrm{s}) + \mathrm{H}_2\,{}^{16}\mathrm{O} \tag{3.47}$$

In this reaction the isotope effect results in enrichment of ^{18}O in precipitated carbonate compared to dissolved carbonate (Fig. 3.5). The equilibrium constant for this reaction and, hence, the $^{18}O/^{16}O$ ratio, can be calculated to have a temperature dependence according to

$$T(^\circ\mathrm{C}) = 18 + 5.44 \times 10^6(\zeta_T - \zeta_{18}) \tag{3.48}$$

where ζ_{18} and ζ_T are the isotopic ratios (cf. (3.4)) at 18°C and at temperature T.

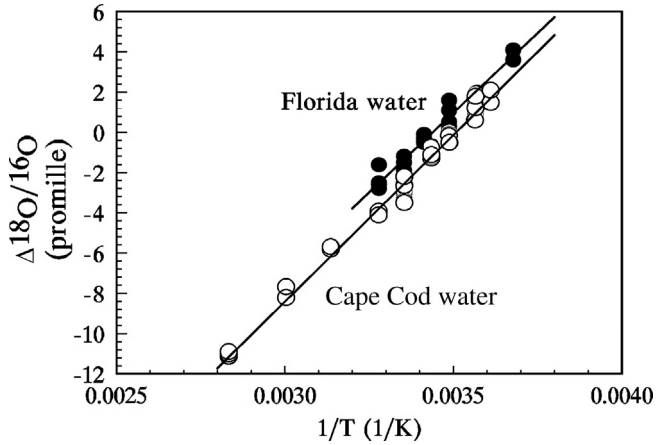


Figure 3.5 Variation of isotopic composition of $CaCO_3$(s) with deposition temperature from water (○-Cape Cod; ●-Florida waters). (From McCrea.)

The isotopic ratios for sedimentary carbonate show a variation of 2.04 to 2.07 $\times$ 10^{-3} (Fig. 3.3). If it is assumed that these differences are due to the precipitation of the carbonate at different temperatures and that the isotopic composition of carbon in seawater was the same as today, one can use the isotopic ratios to obtain information on geologic temperature. In Fig. 3.6 the data for a shell from the Jura geologic period is shown. The oxygen ratio in the carbonate of the shell has been determined by starting from the center of the shell and measuring the isotopic ratio in each layer of annual growth.

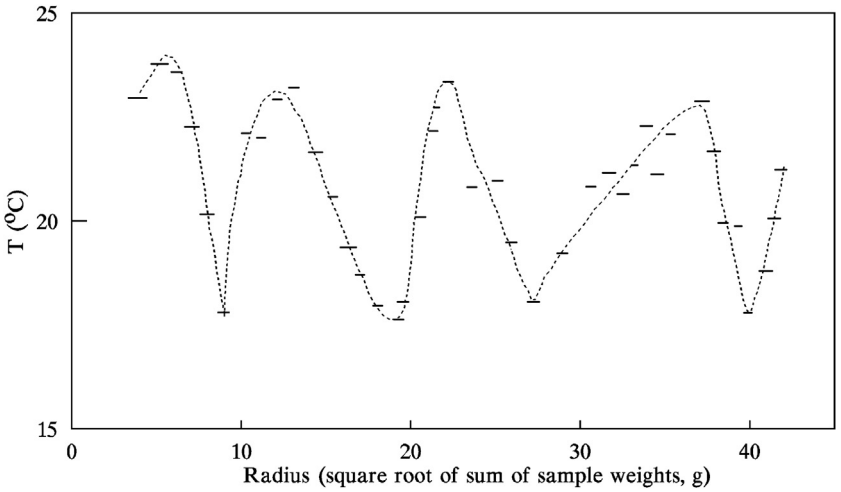


Figure 3.6 Temperature calculated from the $^{18}O/^{16}O$ ratio in carbonate of a shell from the Jura period as function of the distance from center of the shell. (According to Epstein.)

The result shows that the temperature at which the carbonate was formed varied during the life of the creature; further, the creature that inhabited the shell must have died during the spring of its fourth year since there was no further growth of the shell after the forth period of increasing temperature. Thus, by using this method it is possible to determine a small window of yearly temperature variations millions of years back in history.

Chemical isotope effects are particularly large for lighter elements in biological systems. The *chlorella algae* prefers deuterium over hydrogen, and tritium over deuterium. The enrichment factor depends on the conditions of growth; for deuterium to hydrogen an enrichment value of $1.6 - 3$ has been found, while for tritium to hydrogen the enrichment factor is about 2.5. Bacteria behave similarly, e.g. *coli bacteria* showing an enrichment factor for deuterium of 3.9.

In as much as some of the hydrogen atoms are not exchanged readily due to the inertness of their chemical bonds, the isotopic fractionation which involves the easily exchangeable hydrogen atoms in these biological processes must have even larger enrichment factors for deuterium and tritium than their measured values would indicate.

The peculiarity of biological material to prefer certain isotopes has led to studies of how biological material behaves in an isotopic environment which differs substantially from that found in nature. Normally it is found that the organisms wither away and lose their ability to reproduce. Carp cannot survive a higher D_2O concentration than 30%, but, on the other hand, some organisms show a strong growth, and some microorganisms have been found to be able to live in pure D_2O or $H_2{}^{18}O$. It has been possible to raise mice with only ^{13}C in their organism (fed on ^{13}C algae). Exchanging natural $^{14}NH_3$ for $^{15}NH_3$ seems to have little effect on biological systems.

In all of these investigations it should be noted that even when we characterize an isotopic effect as large, it is still quite small by normal reaction criteria except for hydrogen isotopes. As pointed out previously, we can assume that for all but the very lightest elements in most chemical experiments that there is no isotope effect. This assumption forms a basis of the use of radioactive tracers to study chemical systems and diffusion at constant concentration.

3.7. ISOTOPE EFFECTS IN CHEMICAL KINETICS

The reason why higher organisms cannot survive when all light hydrogen atoms are replaced by deuterium is to be found not so much in a shift of chemical equilibria as in a shift in reaction rate leading to a fatal lowering of the metabolic rate when light isotopes are replaced by heavier.

In contrast to chemical equilibria, chemical reaction rates depend on the concentration of the reactants, the transition states and the activation energy but not on the